Finite-branch stalk on the actual weighted-Proj curve

Formal proof packet

27 October 2026

This packet records a conditional geometry proof, not a new target experiment or cost comparison. Its editable diagram, gate CSV, Lean source, and axiom log are beside this PDF.

Checked statement

Let K be a field, let $a_3, a_2, a_1, a_0 \in K$, and let $a \in K$ be a root of the affine quintic f. The new `finiteBranchProjPoint` places the rational branch point in the Z basic open of the actual weighted Proj. The theorem `finiteBranchProjStalkIso` gives an isomorphism from its structure-sheaf stalk to the localization of the affine coordinate ring at the branch ideal.

Actual weighted Proj, Z basic open

$\Downarrow$ checked chart isomorphism and open-immersion stalk maps

$\mathrm{Spec}(A)$ at the rational branch point

$\Downarrow$ checked spectrum-stalk localization

$$\mathcal{O}_{\mathrm{Proj},P} \cong A_{\mathfrak{m}_P}$$

Open: global divisor/Pic^0 equation; compiled Rust law

The open-subspace stalk identification is described in the Stacks Project, Section 26.3. Here the statement is checked against the repository's actual weighted chart in pinned Lean. The earlier local order-one theorem additionally requires a simple root, $N = (x - a)t$, and unit B; this stalk isomorphism does not discharge those hypotheses.

Verification and limits

Direct Lean 4.34.1/Mathlib `d13f23b` compilation exited zero. The theorem's axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Source SHA-256:

`79b790e1ca19b2bacc0c93de87ecb787513717beca77846e55a8db14dfc3994d.`

The theorem does not construct a principal divisor, prove equality in Pic^0, or connect the compiled runtime law to all branches. The previously frozen 1,260-target results, matched-rho controls, and cost ratios remain unchanged.